
Dr. Karanveer Dhingra

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OBJECTIVE:

Interested in a position where all my skills can be put to effective use and take my career to unprecedented heights. I am looking forward to work with my full honesty and dedication and become an asset for the organization.

EXPERIENCE:

- **Associate Professor in** Electrical Engineering Department at CT Institute of Engineering & Technology, Jalandhar. **(20 Sept.2021 to till now)**
- **Assistant Professor** (Guest Faculty) in Electrical Engineering Department at IKGPTU, Main Campus Kapurthala. **(25 Nov. 2020 to 30 July 2021)**
- **Senior Research Fellow** (MANF) at Electrical and Instrumentation Engineering Department, Thapar University, Patiala, India. **(1 April 2017 – 31 March 2020)**
 - *Duties:* To carry out the research work and supervise laboratory exercises of undergraduate students.
- **Junior Research Fellow** (MANF) at Electrical and Instrumentation Engineering Department, Thapar Institute of Engineering & Technology, Patiala, India **(1 April 2015 – 31 March 2017)**
 - *Duties:* To carry out the research work and supervise laboratory exercises of undergraduate students.
- **Assistant Professor** in Electrical Engineering Department at Lovely Professional University, Phagwara. **(19 July 2010 – 29 Nov 2014)**
 - Duties:* To Teach Post graduate and under graduate students
- **Lecturer** at Adesh Polytechnic College, Sri Muktsar Sahib. **(1 Aug 2007- 16 July 2008)**

ACADEMIC QUALIFICATIONS:

- **Ph.D.** “*Frequency Support of Micro-grid Using Fleet of Electric Vehicles*” from Thapar Institute of Engineering & Technology (Deemed University), Patiala, India In **2021**.
- **ME (Power system and Electric Drives)**, Thapar Institute of Engineering & Technology (Deemed University), Patiala, India In **2010** with **9.73 CGPA**.
- **B.Tech (Electrical Engineering)**, Giani Zail Singh College of Engineering and Technology, Bathinda, India in **2007** with **70.65% Marks**.
- **Senior-Secondary (Non-Medical)**, M.S.D Sen. Secondary School, Bathinda, India, In **2003** with **78.22% Marks**.
- **Matriculation**, Tagore Model School, Sri Muktsar Sahib in **2001** with **80.10% Marks**.

Workshops/Short-term Courses Attended:

- Attended workshop on Intellectual Property Rights and Patenting in Thapar Institute of Engineering and Technology on 17th February 2018.
- Attended Short term course on Stability and Control of Renewable Energy based Systems, held from 2nd to 3rd September 2017 in Department of Electrical Engineering, IIT Delhi.
- Attended workshop on MATLAB in Lovely Professional University on 29th June 2013.
- Attended Faculty Development Program on Basic Electrical and Electronics in Lovely Professional University from 11th to 15th June 2012.
- Attended five days Faculty Development Program on “Contemporary Trends in Science and Technology” from 5th April to 9th April 2021.
- Attended five days Faculty Development Program on “Distributed Energy System modelling and Control” from 21st June to 25th June 2021 by AICTE training and learning academy.
- Attended five days Faculty Development Program on “Smart Grid and Sustainable Energy” from 28th June to 2nd July 2021 by AICTE training and learning academy.

Awards and Achievements:

- Merit Certificate Award by Thapar University in 2009.
- Delivered Guest Lecture in NITTTR Chandigarh on the Importance of Electric Vehicles on 10 Jan 2019.
- Junior Research Fellowship during Ph.D. (April 2015 – March 2017); Senior Research Fellowship by University Grants Commission through Maulana Azad National Fellowship (MANF) during Ph.D. (April 2017 – March 2020).

Dissertation:**Ph.D. Thesis: *Frequency Support of Micro-grid using Fleet of EVs:***

Supervisor: Dr. Mukesh Singh (Thapar Institute of Engineering & Technology (Deemed University), India).

Description: Frequency of the Micro-grid was supported by providing energy from the fleet of EVs parked at the Charging station. These EVs behave as an energy storage system as well as load for the system. When there is shortage of power generation at the Micro-grid then EVs behave as a source of power for MG and power is absorbed by the EVs, in the case of surplus power supply

at the MG. This bidirectional flow of flow between vehicles parked at the charging station and MG is controlled through virtual synchronous generator control technique.

M.E. Thesis: *Short Range Fixed Head Multi objective Hydro Thermal Scheduling Using Fuzzy Decision Making Technique*

Supervisor: Ms. Manbir Kaur (Thapar Institute of Engineering & Technology (Deemed University), India.

Description: Hydro thermal scheduling was done using fuzzy decision making technique to minimize the cost of generation as well as to reduce the emission of gases. Coefficients have been found to meet the objectives in optimized way.

Foreign Visit

1. Visited *London, United Kingdom* for presenting paper in *IEEE Conference* in April 2019.

RESEARCH INTERESTS:

- Electric Vehicles
- Micro-grids
- Vehicle to Grid and Grid to vehicle concept

PROFESSIONAL SKILLS:

- Proficiency in
 - Latex
 - MATLAB
 - Opal-RT
 - MS-Office.
- Good communication skills, both oral and written.

No of ME Students Guided:

1. Anubhav Ranjan Tiwari, "Economic Emission Dispatch using fuzzy decision making technique," Lovely Professional University.
2. Aishwarya Sinha, "Power flow analysis of 3 phase distribution networks with distributed generation," Lovely Professional University.
3. Gaddam Siva, "Excitation Control of Synchronous Generator using Fuzzy," Lovely Professional University.

List of (SCI/SCIE) Publications:**Journals**

1. Karanveer Dhingra and Mukesh Singh "*Frequency support in a micro-grid using virtual synchronous generator based charging station,*" IET Renewable Power Generation, vol. 12, no. 9, pp. 1034–1044, Jul. 2018. (IF=3.605)
2. Karanveer Dhingra and Mukesh Singh "*Smart Charging Station to Cater the Sudden Ingress and Egress of EVs while Supporting the Frequency of MG through VSG,*" Arabian Journal for Science and Engineering, Springer, vol 45, pp 6715- 6727. June 2020. (IF=2.230)
3. Karanveer Dhingra and Mukesh Singh "*Handshaking of VSG with Charging Station to Support the Frequency in Microgrid,*" Electrical Engineering, Springer, vol. 102, pp 2349-2362, June 2020 (IF=1.296)

Conference Publications:

1. Karanveer Dhingra and Mukesh Singh "*Comparison of the Virtual Synchronous Generator and Droop Control Techniques to Cater the Unexpected Ingress and Egress of EVs at the CS,*" In International Conference on Automation, Computational and Technology Management (ICACTM), London, United Kingdom, pp. 539-543, IEEE, April 2019.

Non (SCI/SCIE) Publications:

1. Gaddam Siva, Karanveer Dhingra, Priyanka Rani, "*AVR and Fuzzy Logic Power System Stabilizer for the Excitation Control of Synchronous Generator,*" International Journal of Enhanced Research in Science and Technology Engineering, vol. 3, no. 2, pp 61-68.
2. Anubhav Ranjan Tiwari, Karanveer Dhingra, "*Economic Emission Dispatch using Fuzzy Decision making Technique,*" vol. 2, issue no. 4, pp 604-618, April 2014.

REFERENCES:

1. Dr. Mukesh Singh, Associate Professor, EIE Department, Thapar Institute of Engineering & Technology, Patiala, India. E-mail: mukesh.singh@thapar.edu, M. No. +91-9970633998.
2. Dr. Neeraj Kumar, Professor, CSE Department, Thapar Institute of Engineering & Technology, Patiala, India. E-mail: neeraj.kumar@thapar.edu, M. No. +91-8872540189.

3. Dr. Mandeep Singh, Professor, EIE Department, Thapar Institute of Engineering & Technology, Patiala, India. E-mail: mandy_tiet@yahoo.com, M. No. +91- 9855480194
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DECLARATION:

I do hereby declare that the above information is true to the best of my knowledge.

(Karanveer Dhingra)